



Higher-order social networks and order transitions in evacuation: A hypergraph emotion–motion framework

Wenfeng Yi^a, Wenhan Wu^b *,

^a College of Artificial Intelligence, China University of Petroleum, Beijing 102249, China

^b Senseable City Laboratory, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

ARTICLE INFO

Keywords:

Pedestrian dynamics
Higher-order networks
Emotional contagion
Collective behavior
Phase transition

ABSTRACT

With the rapid advancement of computational simulation, crowd evacuation modeling has become a crucial tool for understanding collective human behavior under emergency conditions. However, existing models often fail to capture the complex, non-linear dynamics within human subgroups. From this, we introduce a two-layer framework coupling a social force model with a higher-order emotional contagion mechanism on a simplicial complex to model the threshold-based eruption of collective panic. Supported by empirical comparison, benchmark-based plausibility checks, and mechanism-level controls, our simulations reveal a counter-intuitive finding: subgroups, acting as panic amplifiers through higher-order contagion, make the system less robust by accelerating an order–disorder phase transition. The group panic threshold functions as a critical switch for this collective fragility. Using topological data analysis, we identify a robust negative association between macroscopic order and the network's higher-order connectivity (Hodge Laplacian spectral gap). Our findings highlight the necessity of higher-order models and re-define subgroups as potential panic epicenters, demanding new and group-aware crowd management strategies.

1. Introduction

The efficiency and safety of emergency evacuation during abrupt disasters in crowded venues are of paramount importance [1,2]. However, a long history of stampede incidents reveals that panic-driven crowd evacuations can lead to severe casualties [3,4]. Classic examples include congestion-induced stampedes triggered by panic during fires and sudden disturbances of unknown origin at large-scale events. These incidents underscore the critical need to investigate the mechanisms of evacuation behavior. Despite this, there has been a relative lack of systematic research and quantitative models for collective escape behavior. To mitigate the probability of such disasters and to optimize evacuation strategies [5], a deeper understanding of the interplay between emotional contagion and social relational structures on collective behavior is required [6,7].

In emergency evacuation scenarios, the emotional states of individuals influence each other and propagate across the crowd, this phenomenon is known as emotional contagion. Emotional contagion is the process by which individuals' emotions and behaviors converge through perceiving the emotional changes of others around them [8]. In dense crowds, the spread of panic often triggers a herding effect, causing more individuals to fall into a state of alarm, thereby compromising evacuation efficiency and order [9]. Existing research indicates that the

propagation speed of collective panic is primarily determined by the structural features of the social network, such as network density, the location of the initial panicked individuals, and individual sensitivity to panic [10,11]. In general, a higher average network degree (i.e., denser social contact), a longer distance from the panic source to the exit, and a higher panic propagation coefficient within the group are correlated with lower overall evacuation efficiency [7]. This implies that the rapid spread of emotions can lead to widespread panic, resulting in severe congestion and reduced efficiency. Therefore, integrating emotional contagion mechanisms into evacuation models is crucial for realistically reproducing evacuation dynamics and evaluating intervention strategies [12,13].

Some scholars have begun to focus on the impact of emotional factors on evacuation processes. For instance, prior studies suggested that emotions act as informational cues that bias individual judgment and decision-making during emergencies [8,9]. Numerous models have adapted epidemiological frameworks, such as the Susceptible-Infected-Recovered (SIR) model and its variants, to describe the spread of emotions in escaping crowds [14,15]. These models typically assume that an individual will become panicked if the number of surrounding infected individuals reaches a certain threshold, analogous to the

* Corresponding author.

E-mail address: wenhanwu@mit.edu (W. Wu).

<https://doi.org/10.1016/j.ress.2026.113179>

Received 14 January 2026; Received in revised form 8 May 2026; Accepted 15 July 2026

Available online 21 July 2026

0951-8320/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

threshold effect in epidemiology. Meanwhile, another research has incorporated personality traits to construct more personalized emotional contagion models [16]. Although these models have enhanced the characterization of psychological factors in evacuations, most still remain limited to low-order networks (i.e., networks containing only pairwise interactions) in their representation of interaction structures [17]. In other words, whether based on epidemiological analogies or threshold rules, the fundamental interaction unit is typically assumed to be a dyad. This reliance on simple networks struggles to accurately represent the inherently multi-agent interaction situations that arise in practice. In actual evacuation crowds, social relationships have complex, group-level properties [18]: people often move in subgroups, such as families or friends, where multiple members can jointly and simultaneously influence each other's behavior and emotional state [19–21]. Traditional dyadic networks, which decompose multi-agent influence into a set of pairwise links, fail to capture such synergistic, jointly-reinforcing effects. Prior work on complex contagion has emphasized that certain behavioral or emotional shifts require simultaneous reinforcement from multiple neighbors rather than independent pairwise exposure [12]. Higher-order contagion theory further argues that, for example, when two members of a three-person subgroup display visible panic, the third member's emotional transition is not equivalent to simply summing two independent dyadic influences, it reflects a coordinated three-body signal instead [17].

A recent study has attempted to use higher-order structures (i.e., hypergraphs and simplicial complexes) to characterize these collective phenomena [6]. In higher-order contagion models, social influence can be transmitted via subgroup interactions of three or more individuals, not only along pairwise links [13]. This work reveals several qualitative behaviors that do not appear in standard dyadic network models, including abrupt (discontinuous) transitions at critical thresholds, bistability between low- and high-panic states, hysteresis, and multistability of collective modes [22,23]. These findings indicate that higher-order coupling can fundamentally reshape collective dynamics: panic escalation may occur when a critical mass of panicked subgroup members is reached, and once a high-panic state is established, it can persist even if conditions partially relax. Such non-linear and history-dependent features provide a natural theoretical basis for modeling sudden order–disorder transitions, coordinated surges, and lock-in of panic states during evacuation. Notably, similar mechanisms can also emerge in real human crowds through subgroup formations, and the presence of subgroups significantly affects the spatio-temporal characteristics of pedestrian and evacuation dynamics [24,25]. In real-world evacuations, people who are familiar with each other tend to move together and attempt to maintain cohesion, which introduces structured, quasi-organized flow patterns at the local scale [10,19]. This subgroup behavior alters local congestion patterns and microscopic lane formation: subgroup members align their walking direction, while non-subgroup individuals often yield to intact subgroups moving in formation, producing dynamic avoidance and passage structures. However, most classical evacuation models treat human crowds as a collection of independent individuals and do not explicitly consider these social subgroups [3,26]. This simplification can cause simulated flow patterns and local bottleneck dynamics to deviate from empirical observations, particularly under stress, where subgroup cohesion and mutual waiting behavior become pronounced. A comprehensive representation of real-world dynamics in high-density evacuation scenarios therefore requires incorporating the formation, stability, and interaction mechanisms of subgroups into relevant models [27].

Therefore, this paper introduces a two-layer framework that couples higher-order social structure with a social force model of pedestrian motion. The emotional-contagion dynamics are defined on a subgroup hypergraph, in which each declared social subgroup is represented as a hyperedge. For Hodge-based structural analysis, we construct an associated simplicial complex according to the procedure specified in the Methods section. This distinction is important because a

general hypergraph and a simplicial complex are not interchangeable representations. The proposed framework captures thresholded multi-agent reinforcement while retaining ordinary pairwise proximity and long-range interactions. Through empirical comparison, descriptive plausibility checks, process-level ablation, and pairwise-radius controls, we examine how the tested higher-order mechanism is associated with emotional amplification and order–disorder transitions.

This paper is organized as follows. Section 2 describes a two-layer modeling framework, including the emotionally-coupled social force model and the higher-order emotional contagion mechanism. Section 3 presents extensive numerical experiments, from model validation against empirical data to a systematic parametric study of the order–disorder phase transition, as well as a series of ablation studies. Additional process-level controls are used to distinguish higher-order contagion from ordinary group aggregation and pairwise diffusion. Finally, Section 4 summarizes our key findings and discusses their broader implications and future research directions.

2. Methods

To investigate the coupled mechanism of emotional contagion and pedestrian evacuation under higher-order social network structures, we construct a comprehensive, two-layer model here. This framework, conceptually illustrated in Fig. 1, decouples the agents' physical dynamics from the network of emotional influence. The bottom physical layer describes pedestrian movement in continuous space, governed by the physiological and psychological forces of the social force model (SFM) [3,28]. Concurrently, the top contagion network layer represents the abstract, multi-modal connections through which emotions propagate. The emotional state of each agent in this network is dynamically updated and, in turn, influences their movement parameters in the physical layer.

As shown in Fig. 1, the contagion network is a composite of different connection types: (i) strong, pre-defined higher-order connections that bind subgroups into single computational units for complex contagion; and (ii) pairwise connections that can be short-range (based on spatial proximity and visibility) or long-range (reflecting more distant social ties, akin to small-world networks). The following subsections provide a detailed mathematical and algorithmic description of each layer and their coupling.

2.1. Pedestrian movement model with emotional coupling

The motion of pedestrian i belonging to subgroup G is determined by an extended SFM, where the acceleration is governed by the sum of forces:

$$m_i \frac{d\mathbf{v}_i}{dt} = \mathbf{f}_i^{\text{drive}} + \sum_{j \neq i} \mathbf{f}_{ij}^{\text{rep}} + \sum_w \mathbf{f}_{iw}^{\text{wall}} + \mathbf{f}_i^{\text{group}} \quad (1)$$

where m_i and $\mathbf{v}_i$ are the mass and actual velocity of pedestrian i . The terms on the right side represent the driving force, repulsive forces with other pedestrians and walls, and a cohesive force of subgroup members, respectively. Crucially, the individual's emotional state, denoted by $\theta_i(t)$, directly modulates the primary driving force.

2.1.1. Emotion-driven driving force

The driving force, $\mathbf{f}_i^{\text{drive}}$, models the pedestrian's intention to move towards a target:

$$\mathbf{f}_i^{\text{drive}}(t) = \frac{m_i}{\tau_i} (v_i^{\text{des}}(t) \mathbf{d}_i(t) - \mathbf{v}_i(t)) \quad (2)$$

where τ_i is the relaxation time, representing how quickly the pedestrian adapts the actual velocity. Both the desired speed v_i^{des} and the desired direction $\mathbf{d}_i$ are functions of the instantaneous panic level $\theta_i(t)$ of pedestrian i .

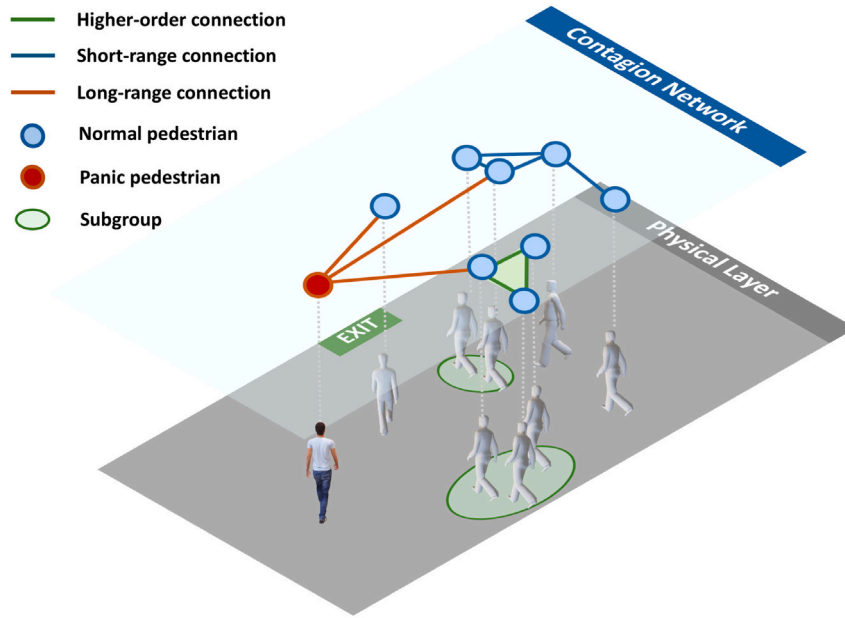


Fig. 1. Conceptual illustration of the coupled two-layer model. The framework consists of a physical layer where pedestrians move according to the social force model, and a contagion network layer where emotional states evolve. The network topology is multi-modal, comprising persistent, higher-order connections within pre-defined subgroups and various pairwise connections. The interplay between these layers governs the overall evacuation dynamics.

First, we assume that a higher panic level leads to a higher desired speed. This is modeled as a linear interpolation between a base velocity v_i^0 (e.g., fast walking) and a maximum velocity $v_i^{\max}$ (sprinting):

$$v_i^{\text{des}}(t) = v_i^0 + (v_i^{\max} - v_i^0) \theta_i(t) \quad (3)$$

This formulation captures the tendency of panicked individuals to run ($\theta_i \rightarrow 1$), while calm individuals move at a slower and more deliberate pace ($\theta_i \rightarrow 0$). Second, the desired direction results from a trade-off between individual escape instinct and subgroup coordination. Two unit vectors are defined: the escape direction $\mathbf{e}_i$ (pointing towards the nearest exit) and the subgroup direction $\mathbf{g}_i$ (pointing towards the centroid of the subgroup). The final desired direction $\mathbf{d}_i$ is a weighted average, with the panic level θ_i serving as the weight [29]:

$$\mathbf{d}_i(t) = \frac{\theta_i(t) \mathbf{e}_i + (1 - \theta_i(t)) \mathbf{g}_i}{\|\theta_i(t) \mathbf{e}_i + (1 - \theta_i(t)) \mathbf{g}_i\|} \quad (4)$$

This implies that calm individuals prioritize group cohesion ($\theta_i \rightarrow 0$), whereas panicked individuals prioritize the most direct escape route, even at the risk of separating from the subgroup ($\theta_i \rightarrow 1$).

2.1.2. Repulsive interaction forces

The remaining forces follow established SFM conventions. The pedestrian-pedestrian repulsive force f_{ij}^{rep} and wall-repulsive force f_{iw}^{wall} are modeled using exponentially decaying potential fields to ensure collision avoidance. The group cohesion force f_{ig}^{group} is an attractive force (e.g., a spring-like force) that activates when a subgroup member strays too far from the group's center, maintaining subgroup integrity. The second term f_{ij}^{rep} is an interaction force with other pedestrians j :

$$\mathbf{f}_{ij}^{\text{rep}} = A_i \exp[(r_{ij} - d_{ij})/B_i] \mathbf{n}_{ij} + k_g(r_{ij} - d_{ij}) \mathbf{n}_{ij} + \kappa_g(r_{ij} - d_{ij}) \Delta v_{ji}^t \mathbf{t}_{ij} \quad (5)$$

where A_i and B_i are constants, d_{ij} is the distance between pedestrians i and j , r_{ij} is the sum of their radii r_i and r_j , and $\mathbf{n}_{ij}$ is the normalized vector pointing from pedestrian j to i . k and κ are body elasticity coefficient and sliding friction coefficient, $\mathbf{t}_{ij}$ is the tangential direction, and Δv_{ji}^t is the tangential velocity difference. Note that $g(x)$ is zero if

pedestrians i and j do not touch each other, otherwise it equals to the argument x .

The wall-interaction term $\mathbf{f}_{iw}^{\text{wall}}$ is a similar interaction force with wall w :

$$\mathbf{f}_{iw}^{\text{wall}} = A_i \exp[(r_i - d_{iw})/B_i] \mathbf{n}_{iw} + k_g(r_i - d_{iw}) \mathbf{n}_{iw} - \kappa_g(r_i - d_{iw}) (\mathbf{v}_i \cdot \mathbf{t}_{iw}) \mathbf{t}_{iw} \quad (6)$$

where d_{iw} is the Euclidean distance to wall w , $\mathbf{n}_{iw}$ is the normalized vector perpendicular to it, and $\mathbf{t}_{iw}$ the direction tangential to it.

2.1.3. Group cohesive force

To maintain the integrity of subgroups, an attractive group cohesive force $\mathbf{f}_i^{\text{group}}$ is introduced. This force acts on pedestrian i as the sum of conditional, spring-like forces from all other members q within the same subgroup G :

$$\mathbf{f}_i^{\text{group}} = \sum_{q \in G, q \neq i} \mathbf{f}_{iq}^{\text{coh}} \quad (7)$$

where the pairwise cohesive force $\mathbf{f}_{iq}^{\text{coh}}$ is defined as:

$$\mathbf{f}_{iq}^{\text{coh}} = \begin{cases} k_G(d_{iq} - d_0) \frac{\mathbf{p}_q - \mathbf{p}_i}{d_{iq}} & \text{if } d_0 < d_{iq} < d_{\max} \\ \mathbf{0} & \text{otherwise} \end{cases} \quad (8)$$

Here, k_G is the cohesion spring constant, and d_0 represents the desired minimum distance between subgroup members. This force is conditional: it only activates within an intermediate range ($d_0, d_{\max}$), pulling straying members back towards the subgroup. The force deactivates if members are too close (i.e., $d_{iq} \leq d_0$), allowing short-range repulsion to dominate, or if they are too far apart (i.e., $d_{iq} \geq d_{\max}$), modeling a lost-contact scenario. This mechanism effectively simulates the tendency of subgroup members to stay together without collapsing into a single point, thus preserving the subgroup's spatial integrity during movement.

2.2. Multi-mechanism emotional contagion

The emotional state of pedestrian i , $\theta_i(t) \in [0, 1]$ (where 0 is completely calm and 1 is maximally panicked), evolves according to a

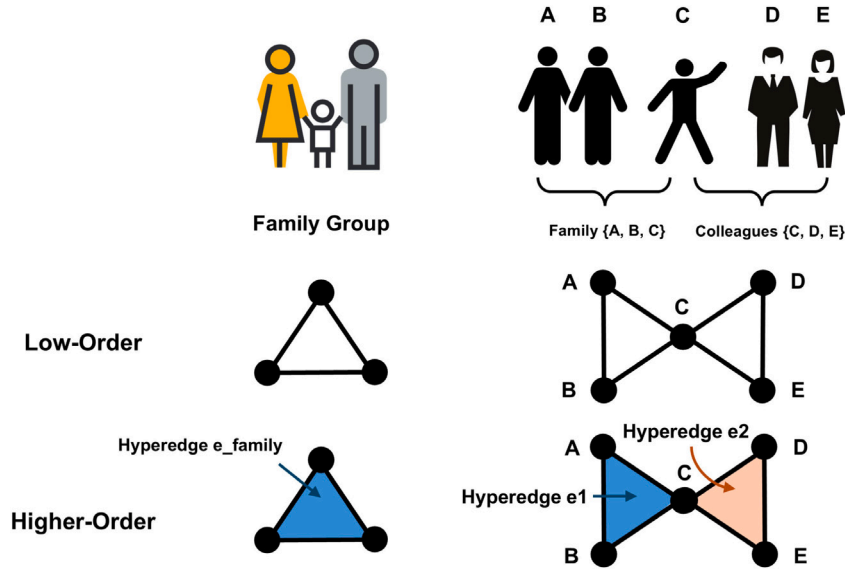


Fig. 2. Comparison of low-order and higher-order network representations for subgroups. Left column: a triadic subgroup is decomposed into three pairwise edges in a low-order network. The higher-order representation captures it as a single, holistic unit (a hyperedge), which is the substrate for the non-linear infection mechanism I_i^{gp} . Right column: for overlapping subgroups (e.g., individual C belongs to a family {A, B, C} and a colleague community {C, D, E}), the higher-order representation with two intersecting hyperedges (e_1 and e_2) explicitly preserves this critical social structure.

continuous dynamical system. The change rate of panic is determined by the interplay of four distinct mechanisms. The overall dynamics are described by the following ordinary differential equation (ODE):

$$\frac{d\theta_i}{dt} = I_i^{\text{nei}}(t) + I_i^{\text{group}}(t) + I_i^{\text{env}}(t) + I_i^{\text{rec}}(t) \quad (9)$$

The variables in this emotional layer have direct behavioral interpretations. The state θ_i is a normalized urgency or panic level that affects both desired speed and the balance between direct escape and subgroup cohesion in the motion layer. The environmental risk $E(t)$ represents background danger cues such as alarms, smoke, or perceived threat intensity. The threshold θ_c determines whether an individual is counted as being in a high-emotion state, while the group panic threshold ϕ determines the fraction of high-emotion members required to trigger subgroup-level amplification. The coefficient η_i controls the strength of this subgroup-triggered amplification. Thus, the emotion model is not an isolated conceptual layer: it is coupled back to observable movement through speed, direction choice, and subgroup cohesion.

2.2.1. Neighbor influence (emotional diffusion)

This term models the diffusion of emotion through pairwise social interactions, driving an individual's emotion towards the local average of their neighbors:

$$I_i^{\text{nei}}(t) = \alpha_i \sum_{j \in \mathcal{N}_i} w_{ij}(\theta_j(t) - \theta_i(t)) \quad (10)$$

Here, $\mathcal{N}_i$ is the set of neighbors of pedestrian i , $\alpha_i > 0$ is the individual's sensitivity coefficient to peer influence, and w_{ij} is a connection weight (e.g., based on inverse distance).

2.2.2. Subgroup infection (higher-order non-linear contagion)

This term captures the core non-linear effect of subgroup panic, modeled as a complex contagion process on the hypergraph structure. An additional emotional boost is triggered only when the fraction of panicked members in a subgroup g exceeds a critical threshold ϕ . The subgroup panic proportion $P_G(t)$ is defined as:

$$P_G(t) = \frac{1}{|G|} \sum_{q \in G} \mathbf{1}_{\theta_q(t) > \theta_c} \quad (11)$$

where $|G|$ is subgroup size, $\mathbf{1}_{\{\cdot\}}$ is the indicator function, and θ_c is a panic threshold defining a panicked state. The infection term is activated via a Heaviside step function $H(\cdot)$:

$$I_i^{\text{group}}(t) = \eta_i H(P_G(t) - \phi)(1 - \theta_i(t)) \quad (12)$$

where $\eta_i > 0$ is the sensitivity to group contagion, and the term $1 - \theta_i(t)$ ensures that calmer individuals are more susceptible to this influence. This mechanism characterizes the sudden outbreak of panic within a subgroup when reaching a tipping point (ϕ). The necessity of representing subgroups as holistic units for this mechanism is illustrated in Fig. 2.

This higher-order term is different from ordinary pairwise diffusion. Pairwise diffusion aggregates independent dyadic exposures, whereas I_i^{group} is activated only after a subgroup-level threshold has been crossed. It therefore represents a collective reinforcement signal from the subgroup as a whole. In the numerical experiments below, we explicitly remove this term and compare it with pairwise-only controls to test whether the observed emotional spread can be explained by ordinary group aggregation alone.

2.2.3. Environmental driver (external field)

This term models the influence of ambient danger cues (e.g., fire, alarms), driven by a global environmental risk function $E(t) \in [0, 1]$:

$$I_i^{\text{env}}(t) = \beta_i E(t)(1 - \theta_i(t)) \quad (13)$$

where $\beta_i > 0$ is the individual sensitivity to environmental threats.

In the present simulations, E is treated as a controlled diagnostic parameter rather than a directly calibrated measurement from a specific real event. This allows us to separate low, moderate, and high background danger conditions. In particular, the process-level ablation analysis uses $E = 0.30$ to avoid saturation by the environmental drive and to expose the contribution of social contagion mechanisms.

2.2.4. Emotional recovery (calm-down decay)

This term represents the natural tendency to calm down in the absence of threatening stimuli, expressed as a linear decay:

$$I_i^{\text{rec}}(t) = -\gamma_i \theta_i(t) \quad (14)$$

where $\gamma_i > 0$ is the calm-down rate of individual i .

2.3. Higher-order network topology analysis

To quantitatively analyze how the social network structure influences the contagion dynamics, several tools from algebraic topology are employed. By representing the social fabric as a simplicial complex S , we compute its Hodge Laplacian and Betti numbers [30,31]. In this construction, pedestrians are represented as 0-simplices, pairwise proximity or subgroup co-membership relations are represented as 1-simplices, and multi-person subgroup relations are represented as higher-order simplices. The complex is updated from the evolving evacuation state, so the topological descriptors summarize the current social-interaction structure rather than a static graph alone. These descriptors are intended to complement ordinary graph measures such as degree, density, connected components, and clustering, because such pairwise measures cannot fully represent multi-agent subgroup reinforcement.

The k th Hodge Laplacian, acting on k -simplices, is defined as:

$$L_k = B_k^T B_k + B_{k+1} B_{k+1}^T \quad (15)$$

where B_k is the k th boundary operator that maps k -simplices to their $(k-1)$ -dimensional boundaries. The spectrum (eigenvalues) of L_k reveals the network connectivity at the level of k -simplices (e.g., nodes for $k=0$, edges for $k=1$, and triangles for $k=2$).

The Betti numbers, $\beta_k = \dim(\ker(L_k))$, are topological invariants that count the number of k -dimensional holes. Here, β_0 counts the number of connected components, β_1 counts the number of independent one-dimensional cycles or loops, representing social circles that could act as bottlenecks for information flow, and β_2 counts the number of two-dimensional voids or cavities, which may indicate segregation between clusters of subgroups.

By analyzing the Betti numbers and the spectral gaps of L_1 and L_2 , we can quantitatively assess the network's capacity for rapid and synchronized emotional contagion. For instance, a network with $\beta_1 = 0$ and a large spectral gap for L_1 is expected to exhibit fast and global synchronization of emotion.

3. Numerical experiments

3.1. Experimental setup

To validate the proposed model and investigate the role of higher-order social structures in pedestrian and evacuation dynamics, we conduct a series of numerical simulations. All simulations are performed within a two-dimensional continuous space representing a rectangular room of size $10\text{ m} \times 6\text{ m}$, with a single exit of width 0.8 m . We focus on a single-exit scenario as it constitutes a fundamental bottleneck, which is ideal for observing congestion and the emergence of order-disorder transitions.

The parameters for our model are divided into two categories: 1) controlling the emotional contagion dynamics and 2) governing the pedestrian movement. The specific values, presented in Tables 1 and 2, are chosen based on the established literature and preliminary simulations. Several emotional-contagion parameters, especially E , η_i , θ_c , and ϕ , are expected to influence the transition behavior. We therefore treat the values in Table 1 as a controlled parameterization for mechanism exploration rather than as universal calibrated constants. To address robustness, we further conduct sensitivity analysis by varying $E \in \{0.30, 0.60, 0.90\}$, $\eta_i \in \{0.10, 0.25, 0.40\}$, $\theta_c \in \{0.50, 0.60, 0.70\}$, and $\phi \in \{0.4, 0.5, 0.6\}$ around the diagnostic baseline.

3.2. Model validation and baseline scenario analysis

Before using our model to explore complex phenomena, it is crucial to assess whether it can reproduce fundamental behaviors observed in real-world evacuations. To this end, we benchmark our simulation

Table 1

Emotional contagion and network parameters.

Symbol	Description	Value
α_i	Sensitivity to peer influence	0.2
r_{prox}	Proximity radius for neighbors ($\mathcal{N}_i$)	1.0 m
η_i	Sensitivity to group contagion	[0.1, 1.0]
θ_c	Panicked state threshold	[0.4, 0.8]
ϕ	Group panic threshold	[0, 1]
β_i	Sensitivity to environment	0.5
γ_i	Calm-down (recovery) rate	0.1
$\theta_i(0)$	Initial emotion (majority)	0.2
—	Initial panic seeds (minority)	5% of pop.

Table 2

Social force model, geometry and numerical settings.

Symbol	Description	Value
μ_v, σ_v	Free-speed distribution	$1.30 \pm 0.30\text{ m s}^{-1}$
m_i	Pedestrian mass	80 kg
r_i	Body radius	[0.15 m, 0.20 m]
τ_i	Relaxation time	0.5 s
A_i	Social repulsion constant	$5 \times 10^2\text{ kg s}^{-2}$
B_i	Range of social repulsion	0.08 m
k	Body elasticity	$1.2 \times 10^5\text{ kg s}^{-2}$
κ	Sliding friction	$2.4 \times 10^5\text{ kg m}^{-1}\text{ s}^{-1}$
c	Queue-alignment gain	$5 \times 10^3\text{ kg m}^{-1}\text{ s}^{-1}$
Δt	Simulation time step	0.04 s

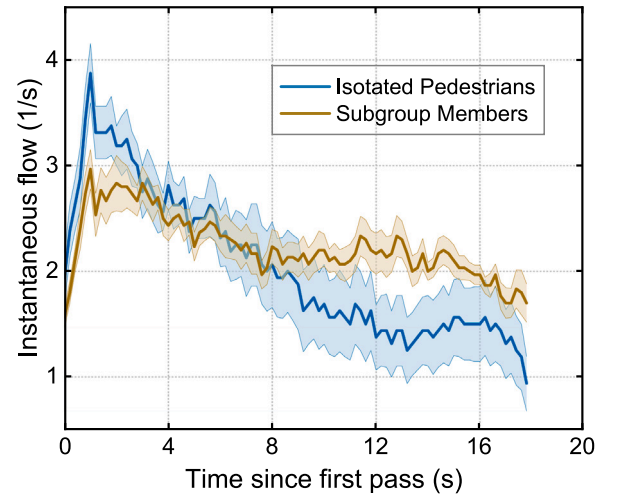


Fig. 3. Instantaneous flow rate from the public experimental dataset. The shaded area represents the standard error across multiple experimental runs.

results against the public experimental dataset [32], which provides detailed trajectories of pedestrians evacuating through a bottleneck, with clear distinctions between individuals belonging to subgroups and those acting alone (isolated).

3.2.1. Analysis of empirical evacuation characteristics

An initial analysis of the empirical data reveals significant behavioral differences between subgroup members and isolated pedestrians. As shown in Fig. 3, we plot the instantaneous flow rate for both categories. The flow of isolated pedestrians (blue curve) exhibits a high initial peak followed by a relatively rapid decay, suggesting an aggressive but less sustainable rush towards the exit. In contrast, the flow of subgroup members (brown curve) has a slightly lower peak but remains more stable and decays more slowly over time. This indicates a more coordinated but less individually optimal evacuation pattern, probably due to members waiting for each other, which moderates the overall flow.

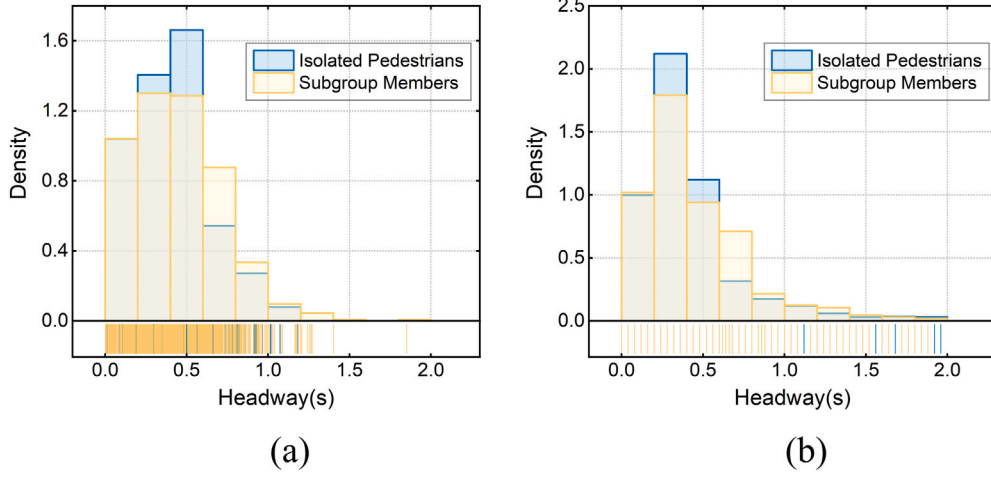


Fig. 4. Comparison of headway distributions between isolated pedestrians and subgroup members. (a) Experimental data. (b) Simulation results.

The macroscopic flow characteristics originate from microscopic differences in passage behavior at the exit. Fig. 4(a) shows the headway distribution (time interval between consecutive passages) for both pedestrian types from the experimental data. The distribution for unaffiliated pedestrians is sharply peaked around 0.5 s, indicative of an efficient, regular passage rhythm. The distribution for subgroup members, however, is broader and has a heavier tail, confirming that their passage is less regular due to the need for intra-subgroup coordination near the exit.

3.2.2. Comparative validation of the model

To assess whether our model can replicate these key empirical findings, we configure a simulation environment to closely match the previous experiment [32], including human crowds with a similar proportion of subgroup members. The headway distribution generated by our model shows strong consistency with the experimental results in Fig. 4(b). Our simulation successfully captures the sharper, narrower distribution for isolated pedestrians and the broader, more dispersed distribution for subgroup members. This comparison supports the ability of the model, which integrates social forces with group cohesion, to reproduce the fundamental micro-dynamics of mixed crowd evacuation. Beyond quantitative metrics, the spatial organization of the crowd near the exit also provides crucial validation. Fig. 5 presents a side-by-side comparison of a snapshot from a real-world experiment and the corresponding snapshot from our simulation. The empirical data, as highlighted by the red circles in Fig. 5(a), clearly shows that pedestrians belonging to subgroups tend to form tight-knit clusters, move together and maintain close physical proximity even in dense environments.

The simulation remarkably reproduces this key qualitative feature. As shown in Fig. 5(b), pedestrians identified as subgroup members (e.g., G1, G2, G3) naturally form cohesive clusters that navigate the bottleneck as a collective unit. Furthermore, our model allows us to visualize the internal states that drive this behavior. The numbers next to each agent represent their panic level θ_i , and it is evident that members within the same subgroup (e.g., G2 and G3, circled in red) exhibit highly similar panic levels ($\theta_i = 0.7$). This homogeneity in emotional state leads to a synchronized decision-making process, where subgroup members adopt similar desired speeds and desired directions, thereby reinforcing their spatial clustering. This provides strong visual evidence that our model's coupling of social cohesion forces with emotional contagion effectively captures the emergent spatial and behavioral patterns of subgroups in evacuations.

Because matched empirical observations of small-group higher-order emotion contagion are not available, we add a benchmark-based quantitative plausibility validation rather than claiming direct

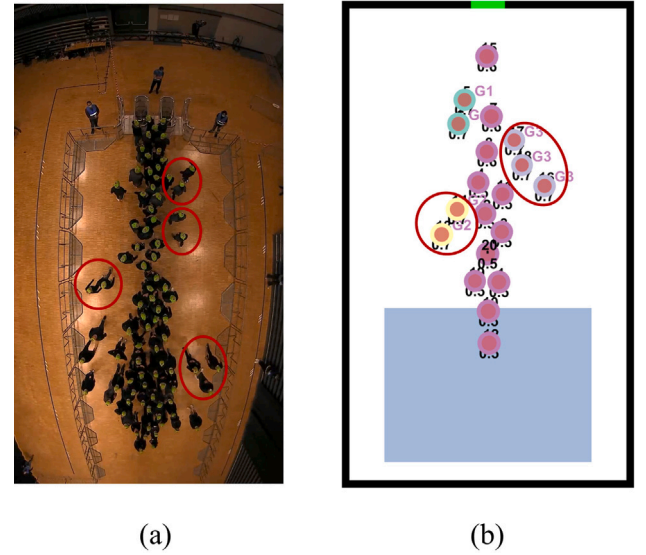


Fig. 5. Visual comparison of queuing behavior at the bottleneck. (a) Snapshot from a real-world crowd experiment [33]. (b) The corresponding snapshot from our simulation.

goodness-of-fit calibration. Table 3 summarizes 20 independent replications for the no-group and all-group settings. The checks cover completion, walking speed, exit flow, headway, overlap, and numerical boundary violations. These diagnostics fall within the expected pedestrian-scale ranges and verify that the simulated evacuation remains physically and numerically plausible before the higher-order contagion mechanism is examined.

3.2.3. Baseline scenario: dynamic evolution of the social network

In addition, the proposed model offers a unique perspective that experimental data cannot provide: a view into the dynamic evolution of abstract social networks during actual evacuation. Fig. 6 illustrates this for a baseline simulation run, where the overall network becomes sparser as the evacuation proceeds and the number of remaining people decreases (dashed blue line). Importantly, the visualization highlights the different nature of the network connections. The short-range (light blue) and long-range (orange) edges are transient, forming and breaking as pedestrians move. In contrast, the intra-subgroup edges (purple) that visualize the higher-order connections remain stable and persistent

Table 3
Benchmark-based quantitative plausibility validation.

Metric	No-group	All-group	Benchmark/check	Result
Completion ratio	1.000 ± 0.000	1.000 ± 0.000	Near-complete evacuation	Pass
Mean walking speed (m/s)	0.669 ± 0.025	0.597 ± 0.034	Pedestrian-scale speed	Pass
Mean exit flow (ped/s)	1.769 ± 0.131	1.684 ± 0.127	Single-exit plausibility	Pass
Mean exit headway (s)	0.606 ± 0.048	0.640 ± 0.059	Bottleneck plausibility	Pass
Pair overlap rate	0.0066 ± 0.0008	0.0081 ± 0.0006	Numerical collision check	Pass
Wall-cell violation rate	0.0000 ± 0.0000	0.0000 ± 0.0000	Hard wall penetration check	Pass
Out-of-bounds rate	0.0000 ± 0.0000	0.0000 ± 0.0000	Domain-boundary check	Pass

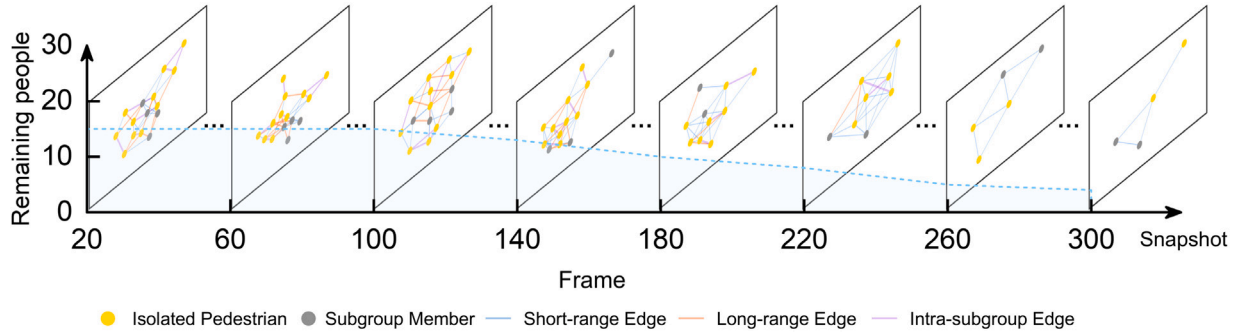


Fig. 6. Dynamic evolution of the social network during a baseline evacuation scenario.

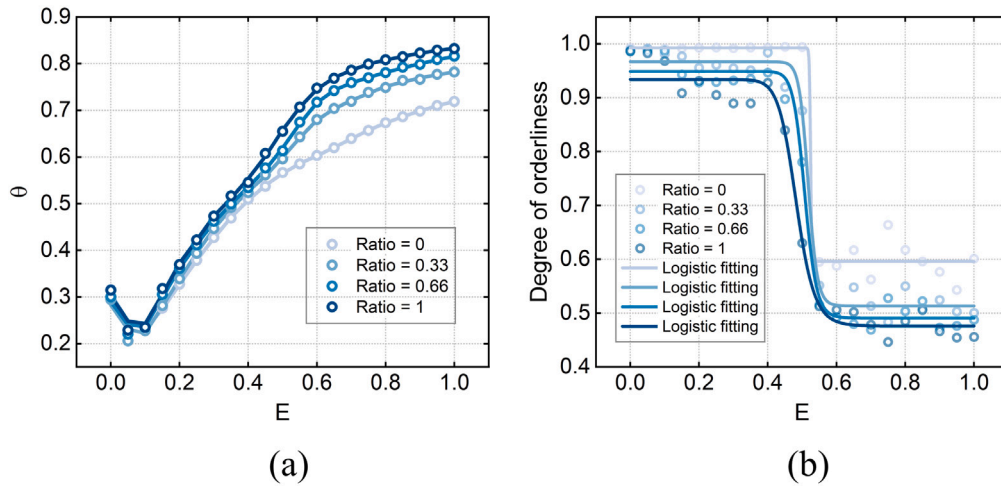


Fig. 7. Effect of subgroup proportion on emotional state and order-disorder transition. (a) Average steady-state panic level $\bar{\theta}$ as a function of the environmental risk E . (b) Degree of orderliness as a function of the environmental risk E .

as long as subgroup members are present in the simulation. This visually demonstrates how pre-defined social relationships act as a durable and underlying structure that continuously affects pedestrian behavior, rather than being a mere emergent property of crowding.

3.3. Parametric study of higher-order effects on order-disorder transitions

After validating the effectiveness of our model, we leverage it to investigate a key research question: how do higher-order social structures and their associated contagion mechanisms influence the emergence of collective order during evacuation? We focus our analysis on the phenomenon characteristic of a dynamical phase transition from an ordered, free-flowing state to a disordered, congested state [29,34].

3.3.1. The role of subgroup proportion in modulating collective behavior

We begin by exploring how the fundamental composition of the crowd (i.e., the proportion of individuals belonging to subgroups) influences the collective emotional state and evacuation order. The subgroup ratio is defined as the fraction of subgroup members in the

total population. Fig. 7(a) shows that, for $E > 0.2$, the average steady-state panic level $\bar{\theta}$ increases monotonically with the subgroup ratio. Thus, a higher prevalence of subgroups amplifies collective panic in our setting. This indicates that the threshold, higher-order contagion within subgroups dominates any potential calming effect of coordinated movement.

This emotional response is reflected in physical order. Here, we define a degree of orderliness as an order parameter based on the polarization of velocities near the exit, where values near 1 indicate smooth, ordered flow, while values close to 0 signify congested, disordered movement. Fig. 7(b) exhibits a sharp transition from ordered to disordered flow as E increases. Importantly, the fitted transition point shifts slightly to lower E as the subgroup proportion grows, implying that larger subgroup prevalence reduces the system's tolerance to environmental risk (i.e., lowers robustness). Contrary to the intuitive expectation that group cohesion might stabilize flow, these results suggest that, under our parameterization, the higher-order contagion mechanism outweighs cohesion, leading to earlier loss of order.

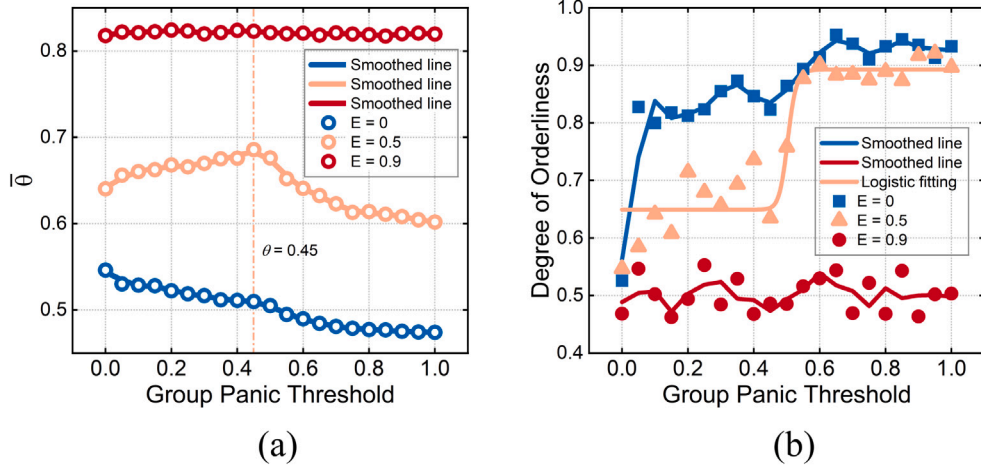


Fig. 8. Group panic threshold (ϕ) as a critical parameter controlling system behavior at low, medium, and high risk levels. (a) Average panic level as a function of the group panic threshold ϕ . (b) Degree of orderliness as a function of the group panic threshold ϕ .

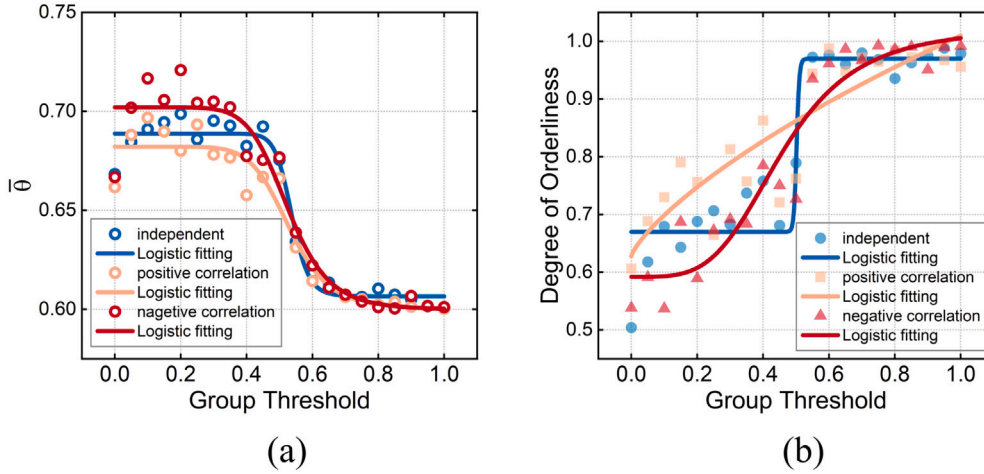


Fig. 9. Effect of correlated heterogeneity between emotional resilience and behavioral cohesion. (a) Average panic level as a function of the group panic threshold ϕ . (b) Degree of orderliness as a function of the group panic threshold ϕ .

3.3.2. Critical role of the higher-order contagion threshold

This part aims to study the specific mechanism of higher-order contagion by analyzing the group panic threshold ϕ . The parameter represents the emotional resilience of subgroups, a higher ϕ indicates that a larger fraction of members must be panicked before triggering a collective emotional outbreak. Fig. 8 shows how the average panic level and orderliness vary with ϕ under three distinct environmental risk levels: low ($E = 0$), medium ($E = 0.5$), and high ($E = 0.9$). At a low risk (blue markers), the system remains calm and ordered, largely insensitive to the group panic threshold. At a high risk (red markers), the system is already in a deep state of panic and disorder, also showing little dependence on ϕ . The most interesting behavior occurs at a medium risk ($E = 0.5$, orange markers). As shown in Fig. 8(a), there is a clear transition point at $\phi \approx 0.45$. The average panic level significantly increases if ϕ is below this critical value (the crowd is emotionally fragile), while it is effectively suppressed once ϕ surpasses this point (the crowd becomes resilient). This emotional transition has a dramatic consequence on the physical order in Fig. 8(b). A sharp transition from a disordered state occurs precisely at this critical threshold. This reveals that the higher-order contagion mechanism acts as a switch, and its threshold ϕ determines whether a moderately risky environment leads to collective order or systemic collapse.

3.3.3. Impact of correlated heterogeneity

In reality, individual traits are often correlated. We explore this by introducing heterogeneity where the group panic threshold ϕ (emotional resilience) is correlated with the group cohesive force (behavioral tendency to stick together). Three scenarios are selected for testing: independent (blue), positive correlation (orange, resilient individuals are more cohesive), and negative correlation (red, resilient individuals are less cohesive). Fig. 9(a) shows that the average panic level exhibits a similar sharp transition at $\phi \approx 0.5$ for all scenarios, suggesting the threshold effect is a robust feature of the higher-order mechanism itself. However, the impact on physical order is profoundly different in Fig. 9(b). For the independent and negatively correlated cases, the transition to an ordered state is sharp and occurs at an intermediate threshold ($\phi \approx 0.5$). In contrast, for the positively correlated case, the transition is much more gradual and delayed. This suggests that strong behavioral bonds act as a stabilizing buffer, preventing the abrupt, system-wide shift seen in other scenarios and leading to a more robust and gradual phase transition.

3.3.4. Topological fingerprints of collective states

To explore the micro-macro link behind the observed phase transition, we analyze the evolving social fabric using the Hodge Laplacian on a simplicial complex representation of the interaction structure [6,

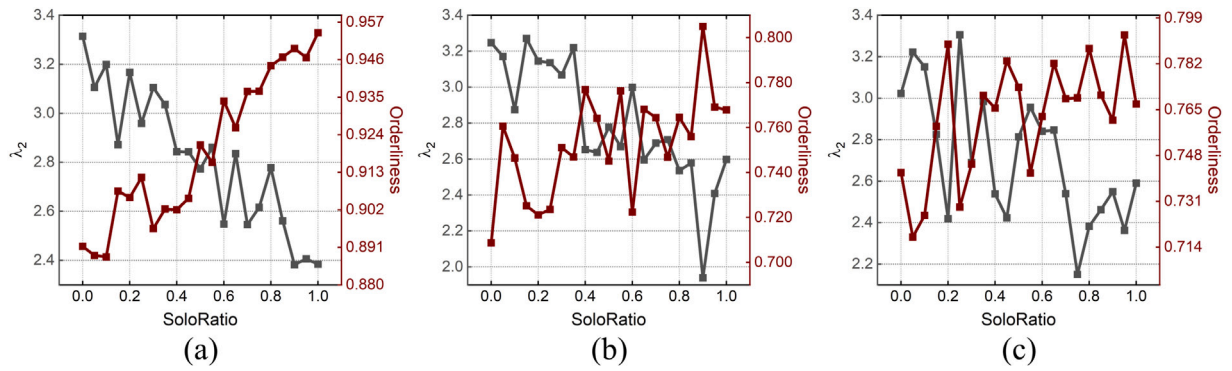


Fig. 10. Topological analysis of the social network. The second spectral gap λ_2 of the Hodge Laplacian and the degree of orderliness as a function of the fraction of isolated pedestrians. (a) Low risk level. (b) Medium risk level. (c) High risk level.

22,35]. Let L_2 denote the second-order Hodge Laplacian acting on 2-simplices (triangles). We are interested in its spectral gap λ_2 , which provides a mesoscopic measure of inter-subgroup connectivity by capturing how tightly different triadic subgroups are coupled through shared faces. Fig. 10 reports λ_2 and the degree of orderliness as the fraction of isolated pedestrians increases (SoloRatio = 1 – GroupRatio) at low, medium, and high risks. A consistent pattern emerges: orderliness increases while λ_2 decreases as SoloRatio grows across all risk levels (i.e., there is a robust negative association between the second-order spectral gap and macroscopic order). The effect is visually strongest at a medium risk in Fig. 10(b), while still discernible at low and high risks in Fig. 10(a) and 10(c).

We interpret this association as a topological fingerprint rather than as a standalone causal proof. The Hodge spectral gap is a mesoscopic descriptor of higher-order connectivity, while other factors such as subgroup ratio, local density, bottleneck geometry, and the strength of emotion-motion coupling can also influence the observed order parameter. Betti numbers, connected components, degree-based measures, and pairwise graph spectra provide complementary but different summaries of the same evolving social fabric. Therefore, the topological analysis is used here to describe how the higher-order social layer changes with crowd composition, and the mechanism-level interpretation is supported by the ablation and pairwise-control experiments reported below. In the present setting, a more tightly connected inter-subgroup topology (larger λ_2) is associated with earlier breakdown of order, whereas a higher share of unaffiliated individuals (smaller λ_2) is associated with more persistent order.

3.4. Disentangling the effects of physical geometry and higher-order social synchronization

The previous sections indicate that higher-order social coordination, particularly fast and group-level synchronization of evacuation intent, can dramatically reshape collective evacuation. A natural question arises whether these effects truly stem from social factors, or whether they can be explained by purely geometric crowding constraints, ordinary physical grouping, or pairwise diffusion alone. To address this, we consider a sequence of complementary analyses. First, we change the egress geometry by introducing an obstacle near the exit and ask whether the link between social topology and macroscopic order survives this perturbation. Second, we compare several behavioral variants of the model that differ in how intent to evacuate is coordinated. Third, we add process-level ablation, pairwise-radius control, and parameter-sensitivity analyses to isolate the higher-order emotional-contagion mechanism more directly.

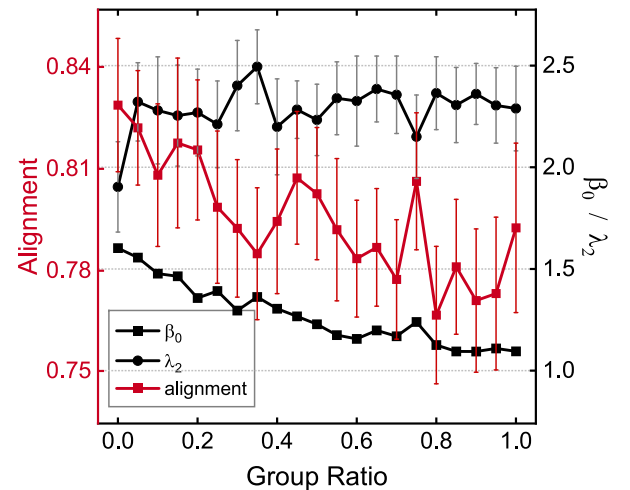


Fig. 11. Topological analysis in the presence of an obstacle. The macroscopic velocity alignment and the topological invariants (β_0 and λ_2) as functions of the group ratio.

3.4.1. Interplay of physical constraints and social topology

We begin by modifying the physical environment: a cylindrical obstacle is placed directly in front of the exit, forcing pedestrians to negotiate a more complex and potentially competitive local geometry. For each crowd composition (parameterized by the fraction of individuals assigned to subgroups, here referred to as the group ratio), we compute both topological descriptors of the effective interaction network and a macroscopic alignment parameter, which is a measure of velocity polarization.

Fig. 11 illustrates three quantities as functions of the group ratio. The black curves show two network-level measures of the interaction graph: β_0 (number of connected components, with lower values indicating less fragmentation) and λ_2 (the second eigenvalue gap, which is a proxy for global connectivity). The red curve shows the corresponding kinematic order (alignment). With the obstacle in place, we observe a clear pattern: as the group ratio increases, β_0 decreases (the graph becomes less fragmented), λ_2 slightly increases (stronger global connectivity), while the alignment tends to decrease. As a result, the alignment does not exhibit an inverse relation with β_0 in our setting. Conversely, the two co-decrease, whereas the alignment is negatively associated with λ_2 . This pattern is consistent with our earlier finding that tighter inter-subgroup coupling (reflected in a larger λ_2) can facilitate higher-order, thresholded co-activation across groups and therefore undermine macroscopic order. The obstacle analysis should be read as a controlled geometry perturbation rather than as proof of robustness across all

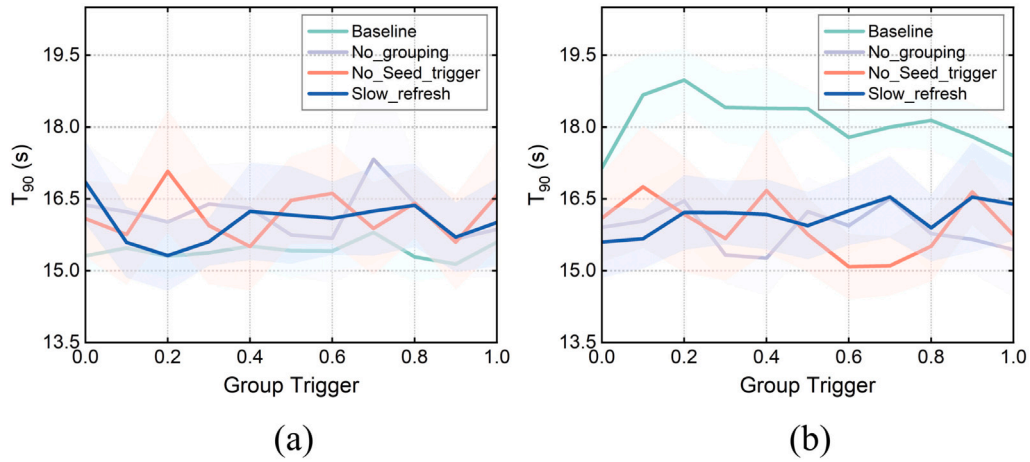


Fig. 12. Ablation on clearance time (T_{90}) as a function of the group-trigger threshold. (a) Crowds composed of isolated pedestrians (group ratio = 0). (b) Crowds composed of full subgroups (group ratio = 1).

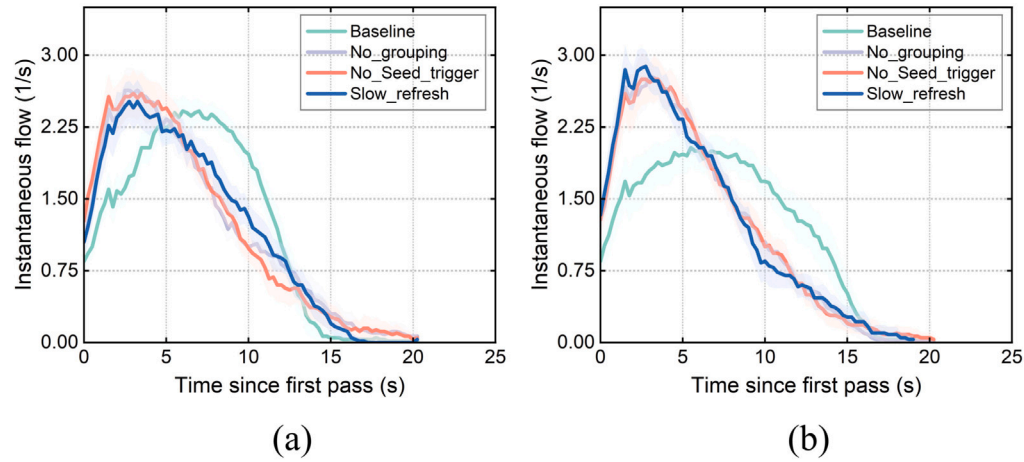


Fig. 13. Instantaneous flow as a function of time since the first egress. (a) Crowds composed of isolated pedestrians (group ratio = 0). (b) Crowds composed of full subgroups (group ratio = 1).

layouts: it suggests that the loss of order is not explained by geometry alone, and that the higher-order social fabric remains an important associated factor in this tested setting.

3.4.2. Collective triggering and evacuation performance

This part is to examine how specific coordination mechanisms affect macroscopic evacuation. We compare four behavioral variants that differ in how evacuation intent is produced and propagated: (i) whether intent can be synchronized at the level of entire social subgroups or only through local pairwise influence; (ii) whether an early, high-urgency seed agent is allowed to trigger the motion of others; and (iii) how quickly intent information is refreshed. For simplicity, the variants are denoted as baseline, no_grouping, no_seed_trigger, and slow_refresh. The efficiency is quantified by T_{90} , the time at which 90% of individuals have exited. Fig. 12 plots T_{90} versus the group-trigger threshold under two limiting compositions. In the case of full subgroups in Fig. 12(b), the synchronized variants achieve shorter T_{90} than the baseline across most trigger values, and the baseline is always the slowest in this scenario. By contrast, when the crowd consists of isolated pedestrians in Fig. 12(a), the separation between variants largely diminishes and the baseline for several trigger ranges is not obviously slower than the alternatives. The macroscopic benefit of rapid social synchronization is conditional: it matters precisely when stable subgroups exist and can act cohesively.

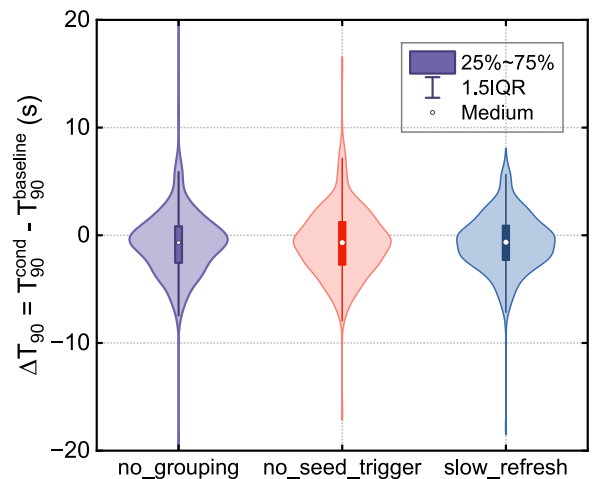


Fig. 14. Violin plots of ΔT_{90} for no_grouping, no_seed_trigger, and slow_refresh relative to the baseline.

To probe the underlying mechanism, we analyze the instantaneous flow at the exit as a function of time since the first egress, averaged over 30 repeated runs. Fig. 13 shows a representative trigger level, where the

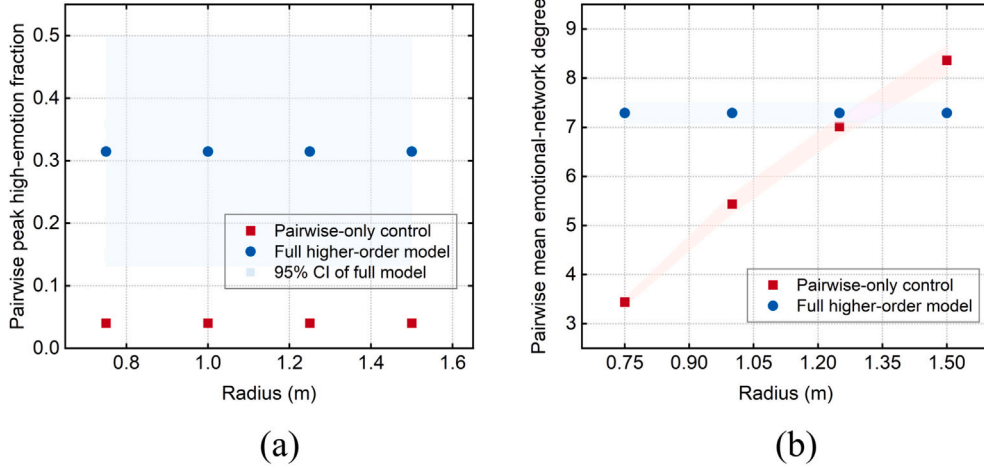


Fig. 15. Process-level ablation of higher-order emotional contagion under $E = 0.30$. (a) Peak fraction of high-emotion pedestrians as a function of the subgroup panic threshold ϕ . (b) Time evolution of the high-emotion fraction at $\phi = 0.5$. (c) Peak number of active higher-order subgroup-triggering events as a function of ϕ . Error bars and shaded bands denote 95% confidence intervals over 20 paired runs.

synchronized variants exhibit an early, high-amplitude peak followed by a rapid decay (pulse-like discharge), but the baseline displays a slower rise with a long tail (sequential, pairwise-mediated release). This contrast is most pronounced when stable subgroups are present. In addition, we also compute $\Delta T_{90} = T_{90}^{\text{cond}} - T_{90}^{\text{baseline}}$ for each initial condition (same layout, trigger, and random seed). Across a large set of paired trials, the distributions of ΔT_{90} for all non-baseline variants are centered below zero in Fig. 14, indicating an average reduction in clearance time relative to the baseline. The negative mean shift is systematic rather than driven by outliers (median < 0 ; inter-quartile ranges well within the negative side). Higher-order social synchronization does more than make individuals decide sooner, because it reshapes the temporal organization of outflow. When cohesive subgroups exist, fast group-level commitment concentrates discharge into an early high-throughput pulse and thins the long tail, thereby reducing T_{90} . In the crowd composed of isolated pedestrians, this advantage largely vanishes, consistent with the view that macroscopic efficiency depends not only on physical bottlenecks but also primarily on the higher-order social topology of the crowd.

3.4.3. Process-level isolation of higher-order emotional contagion

The previous ablation focuses primarily on evacuation performance, whereas clearance time alone is insufficient to identify the specific mechanism driving the observed emotional dynamics. We therefore add a process-level control experiment that retains the same physical grouping while modifying only the emotion layer. The comparison includes the full higher-order model, a no-higher-order control in which the subgroup threshold term is disabled, and a pairwise-only control in which emotional contagion occurs only through local dyadic diffusion. To avoid saturation by the environmental drive, this diagnostic experiment uses $E = 0.30$, where social contagion has enough dynamic range to be identified.

Fig. 15 shows that the higher-order term produces a clear process-level separation. At $\phi = 0.5$, the peak high-emotion fraction is 0.315 ± 0.184 in the full model, but only 0.040 ± 0.000 in both the no-higher-order and pairwise-only controls. The paired full-minus-control difference is 0.275 ± 0.184 over 20 paired runs. The full model also activates subgroup-level triggering events, whereas both controls have zero active higher-order groups by construction. At $\phi = 0.6$, the difference disappears because the subgroup threshold is too high to trigger collective amplification. This threshold dependence supports the interpretation that the observed emotional spread is generated by the higher-order subgroup mechanism rather than by ordinary physical grouping alone.

3.4.4. Pairwise-radius control

We further test whether the pairwise-only control is disadvantaged by an overly narrow contact radius. Holding the physical grouping structure and initial conditions fixed, we disable the higher-order subgroup threshold term and vary only the pairwise emotional-neighbor radius from 0.75 m to 1.50 m. This control directly asks whether ordinary pairwise diffusion can reproduce the full-model emotion spread once the pairwise network is made denser. As shown in Fig. 16, the radius sweep substantially increases pairwise connectivity: the mean emotional-network degree rises from 3.439 ± 0.087 at $r = 0.75$ m to 8.360 ± 0.300 at $r = 1.50$ m. Thus, the pairwise network becomes degree-matched with, and eventually denser than, the full higher-order model (mean degree 7.288 ± 0.214). Nevertheless, the peak high-emotion fraction remains 0.040 ± 0.000 for all tested pairwise radii, while the full higher-order model reaches 0.315 ± 0.184 . This result indicates that the separation between the full model and pairwise controls is not an artifact of choosing a sparse pairwise baseline.

3.4.5. Parameter sensitivity of the emotion-contagion mechanism

To evaluate whether the above conclusions depend on a narrow parameter choice, we perform a one-factor-at-a-time sensitivity analysis around the diagnostic baseline $E = 0.30$, $\eta = 0.25$, and $\theta_c = 0.60$, with $\phi \in \{0.4, 0.5, 0.6\}$. Table 4 summarizes the main observations. The process-level metrics are most sensitive to the environmental drive E and the high-emotion threshold θ_c . Increasing E to 0.60 or 0.90 saturates the high-emotion fraction at 1.0 across the tested thresholds, confirming that strong environmental drive can mask the contribution of social contagion. Increasing θ_c suppresses spread, while lowering it makes subgroup activation easier. The contagion strength η affects amplification after subgroup activation, but its effect is threshold-dependent rather than globally monotonic. Overall, the higher-order mechanism should be interpreted as a threshold-dependent process governed jointly by E , η , θ_c , and ϕ .

4. Conclusions

In this study, we have addressed a critical gap in pedestrian dynamics modeling: the oversimplification of social interactions, which often neglects the profound influence of subgroup structures and collective emotional contagion on evacuation efficiency and safety. To this end, we propose a novel, two-layer model that integrates a social force model with a sophisticated emotional contagion mechanism operating on a higher-order social network, which is a phenomenon termed

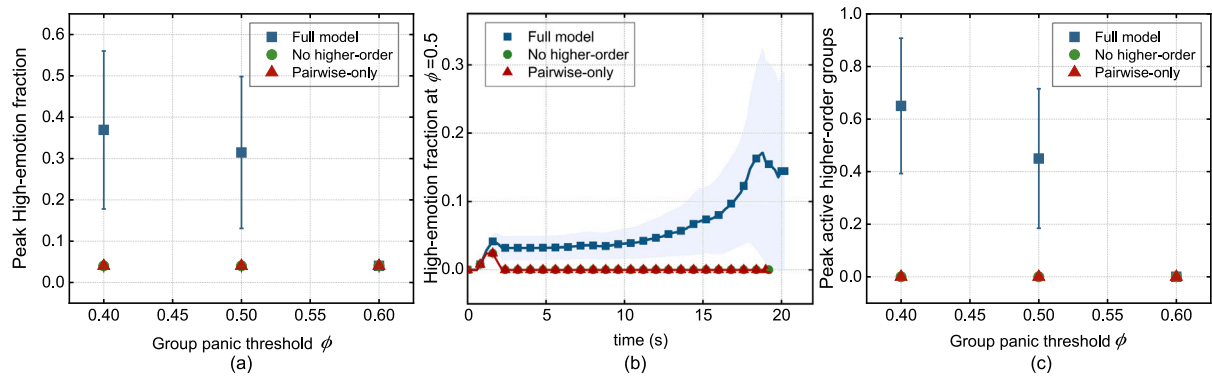


Fig. 16. Pairwise-radius control for ordinary pairwise emotional contagion under $E = 0.30$ and $\phi = 0.5$. (a) Peak fraction of high-emotion pedestrians under pairwise-only contagion as the pairwise emotional-neighbor radius increases from 0.75 m to 1.50 m. (b) Mean emotional-network degree under the same radius sweep. The horizontal reference line and shaded band indicate the mean and 95% confidence interval of the full higher-order model. Error bars denote 95% confidence intervals over 20 paired runs.

Table 4

Summary of parameter sensitivity analysis.

Parameter varied	Range	Main observation
Environmental drive E	0.30, 0.60, 0.90	Higher E saturates emotion spread; at $\phi = 0.5$, peak high-emotion fraction increases from 0.315 ± 0.184 at $E = 0.30$ to 1.000 ± 0.000 at $E = 0.60$ and $E = 0.90$.
High-emotion threshold θ_c	0.50, 0.60, 0.70	Larger θ_c suppresses spread; at $\phi = 0.5$, the peak high-emotion fraction changes from 0.424 ± 0.211 to 0.315 ± 0.184 and then to 0.088 ± 0.094 .
Contagion strength η	0.10, 0.25, 0.40	η controls amplification after subgroup activation; at $\phi = 0.5$, $\eta = 0.10$ remains close to the seeded fraction, whereas $\eta = 0.25$ and 0.40 allow subgroup-triggered spread.
Group threshold ϕ	0.4, 0.5, 0.6	The mechanism remains threshold-dependent: $\phi = 0.6$ suppresses activation under $E = 0.30$, while $\phi = 0.4$ and 0.5 allow contagion in a subset of paired runs.

higher-order emotional contagion. Our comprehensive simulation experiments yield several key findings that together form a coherent narrative. First, our model is supported by empirical comparison and benchmark-based plausibility validation. Second, we find that the collective behavior of the crowd exhibits a critical order–disorder phase transition as the environmental risk increases. Most significantly, and contrary to common intuition, we demonstrate that the presence of subgroups can make the system less robust by shifting this transition to lower risk levels. This key finding reveals that the higher-order emotional contagion mechanism can act as a powerful amplifier for panic, an effect that can dominate the potentially stabilizing forces of group cohesion. The new process-level controls further show that this amplification is not reproduced when the subgroup threshold term is removed or replaced by degree-matched pairwise diffusion.

The implications of our findings are twofold. For the field of complex systems science, this work underscores the necessity of incorporating higher-order interaction models to understand collective phenomena, as they can fundamentally alter system-level dynamics in ways that pairwise models cannot capture [6,7,35]. For practical applications in crowd management and architectural design, our results suggest a paradigm shift away from treating crowds as homogeneous masses of individuals [36,37]. Recognizing the social fabric of a crowd is crucial, but our findings call for a more nuanced approach than simply preserving subgroup integrity [4,38]. Crucially, instead of leveraging an assumed inherent stability, emergency protocols must account for subgroups as potential epicenters of rapid, cascading panic. This suggests that strategies should focus on mitigating intra-subgroup contagion, for example, through targeted communication, to prevent a localized panic from triggering a systemic collapse [39]. More specifically, the threshold results suggest that group-aware interventions can be interpreted as attempts to increase the effective group panic threshold ϕ or to weaken inter-subgroup emotional coupling near bottlenecks. In the present simulations, disorder was suppressed when the effective subgroup threshold exceeded approximately 0.45–0.50 under moderate

risk. This does not imply a universal design code, but it provides a model-scale target for strategies such as clear subgroup-level announcements, trained staff guidance, staged release of dense clusters, lane guidance near exits, and spatial arrangements that reduce unnecessary coupling between adjacent subgroups. Such strategies are most relevant to venues where stable small groups are common, including stadiums, shopping malls, schools, and transport hubs.

Apart from these contributions, our work also opens several avenues for future research. The simulations presented here focus on non-overlapping subgroup structures, and future work should explore more complex and realistic social topologies, such as overlapping and nested communities, which can introduce novel dynamical effects [7,35]. The current numerical scenarios are also limited to a controlled 10 m $\times$ 6 m single-exit setting with an additional obstacle test. This setting is appropriate for mechanism identification, but it is not sufficient to establish venue-general engineering prescriptions. Future work should test multi-exit layouts, obstacle arrays, corridor-room transitions, heterogeneous exit visibility, and calibrated venue-specific data when available. Furthermore, the emotional model could be enriched with a wider spectrum of states beyond panic and calm, and individual heterogeneity could be more deeply explored by linking model parameters to established personality traits [27]. By continuing to bridge the gap between microscopic social mechanisms and macroscopic collective behavior, we can move towards a more predictive and human-centric science of crowd dynamics.

CRedit authorship contribution statement

Wenfeng Yi: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Funding acquisition, Writing – review & editing. **Wenhao Wu:** Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Conceptualization, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 62506386).

Data availability

Data will be made available on request.

References

- [1] Feng X, Gai W. Extended reality technology applied to emergency evacuation research, training and on-site guidance: Analysis and future research. *Reliab Eng Syst Saf* 2026;268:112022. <http://dx.doi.org/10.1016/j.res.2025.112022>.
- [2] Yang S, Ceferino L, Zhang Y, Gu C, Guo T, Kondo G. When agents learn to think: Large language model-enhanced agent-based modeling for crowd evacuation in disaster scenarios. *Reliab Eng Syst Saf* 2026;269:112056. <http://dx.doi.org/10.1016/j.res.2025.112056>.
- [3] Helbing D, Farkas I, Vicsek T. Simulating dynamical features of escape panic. *Nature* 2000;407(6803):487–90. <http://dx.doi.org/10.1038/35035023>.
- [4] Haghani M, Lovreglio R. Data-based tools can prevent crowd crushes. *Science* 2022;378(6624):1060–1. <http://dx.doi.org/10.1126/science.adf5949>.
- [5] Yi W, Wu W. Control strategies for order–disorder phase transition in crowd evacuation. *Reliab Eng Syst Saf* 2026;266:111688. <http://dx.doi.org/10.1016/j.res.2025.111688>.
- [6] Battiston F, Amico E, Barrat A, Bianconi G, Ferraz de Arruda G, Franceschiello B, Iacopini I, Kéfi S, Latora V, Moreno Y, Murray MM, Peixoto TP, Vaccarino F, Petri G. The physics of higher-order interactions in complex systems. *Nat Phys* 2021;17(10):1093–8. <http://dx.doi.org/10.1038/s41567-021-01371-4>.
- [7] St-Onge G, Iacopini I, Latora V, Barrat A, Petri G, Allard A, Hébert-Dufresne L. Influential groups for seeding and sustaining nonlinear contagion in heterogeneous hypergraphs. *Commun Phys* 2022;5(1). <http://dx.doi.org/10.1038/s42005-021-00788-w>.
- [8] Haghani M, Sarvi M. ‘Herding’ in direction choice-making during collective escape of crowds: How likely is it and what moderates it? *Saf Sci* 2019;115:362–75. <http://dx.doi.org/10.1016/j.ssci.2019.02.034>.
- [9] Haghani M, Sarvi M. Imitative (herd) behaviour in direction decision-making hinders efficiency of crowd evacuation processes. *Saf Sci* 2019;114:49–60. <http://dx.doi.org/10.1016/j.ssci.2018.12.026>.
- [10] Haghani M, Sarvi M, Shahhoseini Z, Boltes M. Dynamics of social groups’ decision-making in evacuations. *Transp Res Part C: Emerg Technol* 2019;104:135–57. <http://dx.doi.org/10.1016/j.trc.2019.04.029>.
- [11] Haghani M, Sarvi M. Simulating dynamics of adaptive exit-choice changing in crowd evacuations: Model implementation and behavioural interpretations. *Transp Res Part C: Emerg Technol* 2019;103:56–82. <http://dx.doi.org/10.1016/j.trc.2019.04.009>.
- [12] Ferraz de Arruda G, Petri G, Rodriguez PM, Moreno Y. Multistability, intermittency, and hybrid transitions in social contagion models on hypergraphs. *Nat Commun* 2023;14(1). <http://dx.doi.org/10.1038/s41467-023-37118-3>.
- [13] de Arruda GF, Petri G, Moreno Y. Social contagion models on hypergraphs. *Phys Rev Res* 2020;2(2):023032. <http://dx.doi.org/10.1103/physrevresearch.2.023032>.
- [14] Fu L, Song W, Lv W, Lo S. Simulation of emotional contagion using modified SIR model: A cellular automaton approach. *Phys A* 2014;405:380–91. <http://dx.doi.org/10.1016/j.physa.2014.03.043>.
- [15] Agnelli JP, Buffa B, Knopoff D, Torres G. A spatial kinetic model of crowd evacuation dynamics with infectious disease contagion. *Bull Math Biol* 2023;85(4). <http://dx.doi.org/10.1007/s11538-023-01127-6>.
- [16] Zheng Z, Zhu G, Sun Z, Wang Z, Li L. Improved social force model based on emotional contagion and evacuation assistant. *IEEE Access* 2020;8:195989–6001. <http://dx.doi.org/10.1109/access.2020.3034348>.
- [17] Iacopini I, Petri G, Barrat A, Latora V. Simplicial models of social contagion. *Nat Commun* 2019;10(1). <http://dx.doi.org/10.1038/s41467-019-10431-6>.
- [18] Wu W, Zheng X. A systematic analysis of subgroup research in pedestrian and evacuation dynamics. *IEEE Trans Intell Transp Syst* 2024;25(2):1225–46. <http://dx.doi.org/10.1109/tits.2023.3318417>.
- [19] Xie W, Lee EWM, Li T, Shi M, Cao R, Zhang Y. A study of group effects in pedestrian crowd evacuation: Experiments, modelling and simulation. *Saf Sci* 2021;133:105029. <http://dx.doi.org/10.1016/j.ssci.2020.105029>.
- [20] Wu W, Yi W, Wang X, Wang E, Zheng X. Experimental study on the decision-making and motion behavior of subgroups when facing a static obstacle during movement. *Expert Syst Appl* 2024;242:122761. <http://dx.doi.org/10.1016/j.eswa.2023.122761>.
- [21] Wu W, Yi W. Modeling the dynamical process of behavioral contagion in human crowds during evacuation. *Reliab Eng Syst Saf* 2026;266:111649. <http://dx.doi.org/10.1016/j.res.2025.111649>.
- [22] Gambuzza LV, Di Patti F, Gallo L, Lepri S, Romance M, Criado R, Frasca M, Latora V, Boccaletti S. Stability of synchronization in simplicial complexes. *Nat Commun* 2021;12(1). <http://dx.doi.org/10.1038/s41467-021-21486-9>.
- [23] Iacopini I, Petri G, Baronchelli A, Barrat A. Group interactions modulate critical mass dynamics in social convention. *Commun Phys* 2022;5(1). <http://dx.doi.org/10.1038/s42005-022-00845-y>.
- [24] Wu W, Yi W, Wang X, Zheng X. A force-based model for adaptively controlling the spatial configuration of pedestrian subgroups at non-extreme densities. *Transp Res Part C: Emerg Technol* 2023;152:104154. <http://dx.doi.org/10.1016/j.trc.2023.104154>.
- [25] Wu W, Yi W, Wang X, Zheng X. A vision-driven model based on cognitive heuristics for simulating subgroup behaviors during evacuation. *IEEE Trans Intell Transp Syst* 2024;25(11):16048–58. <http://dx.doi.org/10.1109/tits.2024.3421626>.
- [26] Haghani M, Sarvi M, Shahhoseini Z. When ‘push’ does not come to ‘shove’: Revisiting ‘faster is slower’ in collective egress of human crowds. *Transp Res Part A: Policy Pr* 2019;122:51–69. <http://dx.doi.org/10.1016/j.tra.2019.02.007>.
- [27] Haghani M, Yazdani M. How simple behavioural modifications can influence evacuation efficiency of crowds: Part 2. Physical movement of individuals. *Transp Res Part C: Emerg Technol* 2024;166:104762. <http://dx.doi.org/10.1016/j.trc.2024.104762>.
- [28] Helbing D, Molnár P. Social force model for pedestrian dynamics. *Phys Rev E* 1995;51(5):4282–6. <http://dx.doi.org/10.1103/physreve.51.4282>.
- [29] Yi W, Wu W, Wang X, Wang E, Zheng X. Order–disorder phase transitions in front of the exit during human crowd evacuations. *Transp Res Part C: Emerg Technol* 2024;163:104649. <http://dx.doi.org/10.1016/j.trc.2024.104649>.
- [30] Bick C, Gross E, Harrington HA, Schaub MT. What are higher-order networks? *SIAM Rev* 2023;65(3):686–731. <http://dx.doi.org/10.1137/21m1414024>.
- [31] Schaub MT, Benson AR, Horn P, Lippner G, Jadbabaie A. Random walks on simplicial complexes and the normalized hodge 1-Laplacian. *SIAM Rev* 2020;62(2):353–91. <http://dx.doi.org/10.1137/18m1201019>.
- [32] von Krüchten C, Schadschneider A. Empirical study on social groups in pedestrian evacuation dynamics. *Phys A* 2017;475:129–41. <http://dx.doi.org/10.1016/j.physa.2017.02.004>.
- [33] Sieben A, Postmes T. Behavioural repertoires in moving crowds: An observational approach. *R Soc Open Sci* 2025;12(2). <http://dx.doi.org/10.1098/rsos.241561>.
- [34] Yi W, Wu W, Wang X, Zheng X. Phase transitions in pedestrian evacuation: A dynamic modeling with small-world networks. *IEEE Trans Intell Transp Syst* 2024;25(11):18025–37. <http://dx.doi.org/10.1109/tits.2024.3433420>.
- [35] Zhang Y, Lucas M, Battiston F. Higher-order interactions shape collective dynamics differently in hypergraphs and simplicial complexes. *Nat Commun* 2023;14(1). <http://dx.doi.org/10.1038/s41467-023-37190-9>.
- [36] Wu W, Chen M, Li J, Liu B, Zheng X. An extended social force model via pedestrian heterogeneity affecting the self-driven force. *IEEE Trans Intell Transp Syst* 2022;23(7):7974–86. <http://dx.doi.org/10.1109/tits.2021.3074914>.
- [37] Wu W, Li J, Yi W, Zheng X. Modeling crowd evacuation via behavioural heterogeneity-based social force model. *IEEE Trans Intell Transp Syst* 2022;23(9):15476–86. <http://dx.doi.org/10.1109/tits.2022.3140823>.
- [38] Drury J. The psychology of crowd behavior. *Annu Rev Psychol* 2025. <http://dx.doi.org/10.1146/annurev-psych-012125-121447>.
- [39] Haghani M, Yazdani M. How behavioural changes in social groups affect evacuation efficiency of crowds. *Saf Sci* 2025;181:106679. <http://dx.doi.org/10.1016/j.ssci.2024.106679>.